

A Hybrid AI Reasoner for Forensic Occupancy Detection in Smart Environments

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Abstract

The increasing presence of IoT devices has transformed buildings into smart environments, where sensors continuously record contextual information. Pure machine learning methods struggle to generalise beyond the training data, which motivates the exploration of hybrid approaches that combine the efficiency of machine learning models with the adaptability of symbolic reasoning. This paper develops a hybrid architecture where machine learning components extract low-level events (e.g., room occupancy, sensor anomalies), and a Belief-Desire-Intention (BDI) agent reasons over them to identify room occupancy in smart environments. We empirically evaluate an implementation of the resulting architecture in a forensics scenario under increasingly degraded sensor conditions, showcasing its robustness.

1 Introduction

In forensic investigations, smart environments offer valuable opportunities to reconstruct events by analysing occupancy patterns and human presence across time and space Dey *et al.* [2016]. Determining the number of individuals present in a given location at a specific time is often crucial for validating or refuting alibis, narrowing down suspects, and establishing causal links in criminal cases Singh *et al.* [2018]. However, relying on cameras and sensors to provide this information is far from trivial. In real-world scenarios, devices may be disabled, malfunctioning, or absent due to privacy constraints, technical limitations, or attacks. These gaps create significant challenges for criminal investigators, as incomplete or inconsistent data can compromise the reconstruction of events and prevent the production of reliable evidence Bengtsson [2025]; Stoyanova *et al.* [2020]; Ahmed *et al.* [2024].

Artificial Intelligence provides powerful tools to address some of these challenges. However, traditional Machine Learning (ML) approaches reveal important limitations, such as a lack of transparency and generalisation, which are essential in digital forensic environments. Furthermore, they typically require complete and noise-free input, making them fragile in forensic contexts where missing or corrupted data is common Okafor and Delaney [2021].

In contrast, symbolic AI relies on explicit logical rules, structured representations, and formal inference systems. The Belief-Desire-Intention (BDI) architecture is an example of symbolic AI inspired by human cognitive processes modelled using three key mental states: beliefs, desires, and intentions Bratman *et al.* [1988]. BDI agents provide a useful abstraction to design systems that can reason, plan, and act autonomously in complex environments de Silva *et al.* [2020]; Cardoso and Ferrando [2021]. However, manually creating such plans to cover all possible situations in complex real-world environments is a laborious process. These issues motivate the exploration of hybrid AI approaches, which combine the adaptability of ML models with symbolic reasoning Sarker *et al.* [2021].

In this work, we develop a hybrid AI reasoner to support forensic investigations in smart buildings. The ML component is responsible for perception tasks, such as inferring occupancy from Internet of Things (IoT) sensor data streams. Additionally, a symbolic BDI agent intervenes when the model's confidence is low, applying rule-based reasoning to determine occupancy.

The contributions of this paper are: (1) to investigate how forensic analyses and predictions can be improved under realistic conditions of missing or unreliable sensor data; (2) to explore strategies for adopting a hybrid AI system in digital forensics; and (3) to evaluate the flexibility of the proposed approach across different scenarios.

2 Background

Digital forensics is a scientific discipline focused on the collection, analysis, and preservation of digital data (data produced by electronic devices such as computers, mobile phones, and IoT devices) for use as evidence in legal proceedings Ahmed *et al.* [2024]. It is an area that increasingly depends on rigorous experimentation to validate the reliability and admissibility of evidence.

More recently, IoT ecosystems have expanded both the scope and complexity of forensic investigations. IoT devices generate continuous, multimodal streams of evidence but also introduce novel vulnerabilities. Anti-forensics techniques—such as evidence tampering, obfuscation, and deliberate device disruption—are increasingly observed in IoT attacks, raising the stakes for investigators. New approaches must

therefore integrate advanced forensic strategies to ensure evidential integrity in such environments Yaacoub *et al.* [2022].

In built/IoT environments, supervised ML models trained on multivariate sensor data (e.g., temperature, CO₂, light, and sound) are widely used to infer occupancy or behaviours that contextualise events Dey *et al.* [2016]; Singh *et al.* [2018]. Smart-home and building studies show both the value and the fragility of sensor fusion. Models must tolerate missing data, drift, and device diversity while remaining resilient to tampering and spoofing Yaacoub *et al.* [2022]; Bengtsson [2025].

Because forensic conclusions must be auditable and defensible, there is growing emphasis on interpretable (“glass-box”) approaches and on hybrid designs that combine statistical learning with explicit knowledge Garrett and Rudin [2023]; Costantini *et al.* [2019]; Sikos [2021]; Grojek and Sikos [2022]. Evidence from safety-critical and security domains suggests that symbolic agents or constraints can complement ML, improving robustness and decision transparency Al Shukairi and Cardoso [2023]; Grov *et al.* [2024]; Jalaian and Bastian [2023]; Raza *et al.* [2024]; Delvecchio *et al.* [2025]. Broader surveys position neuro-symbolic AI as a principled path to unify perception and reasoning—aligning with reproducibility and reporting practices expected in digital forensics Sarker *et al.* [2021]; OliveiraJr *et al.* [2020]; Arshad *et al.* [2018].

3 Related Work

This section reviews previous research on the application of AI techniques in digital forensics. ML techniques have gained significant traction in digital forensics due to their ability to model high-dimensional, heterogeneous, and noisy data. These approaches are particularly effective in tasks such as anomaly detection, evidence triage, pattern recognition, and behavioural inference.

Deconto *et al.* [2024] employs supervised learning algorithms to infer human occupancy patterns from IoT sensor data in smart environments. Their approach enables the identification of anomalous or potentially suspicious behaviour, aiding in context-aware forensic reasoning. In the domain of mobile forensics, Barmpatsalou *et al.* [2018] propose fuzzy neural networks to detect irregular communication behaviours, showcasing their utility in mobile threat detection. Henseler *et al.* [2013] focuses on pattern discovery across disparate digital evidence sources, introducing ML algorithms that can correlate behaviour across time and device contexts, thereby aiding suspect profiling. Similarly, Shahzad *et al.* [2020] outlines a comprehensive framework that incorporates deep learning, natural language processing, and anomaly detection to accelerate digital investigations and manage large volumes of unstructured evidence.

Despite their empirical success, ML-based forensic systems remain challenged by a lack of explainability and generalisation OliveiraJr *et al.* [2020]; Arshad *et al.* [2018]; Sarker *et al.* [2021]; Jan and Hussain [2025]. The opaque nature of deep models inhibits transparency, raising serious concerns regarding legal admissibility, investigator trust, and post hoc auditability of evidence Solanke [2022]. Furthermore, deep learning models suffer from the generalisation problem,

which can lead to poor performance when applied to unfamiliar inputs (e.g., different rooms, new sensors, or faulty sensors). These limitations underscore the need for hybrid AI frameworks in digital forensic environments.

Symbolic AI techniques often rely on logic-based reasoning, ontologies, and rule-based inference systems. They are highly valued in forensics for their transparency, interpretability, and alignment with legal and investigative standards. For example, Costantini *et al.* [2019] demonstrates the use of Answer Set Programming (ASP) to model investigative reasoning, enabling investigators to explicitly encode domain knowledge and legal constraints. By contrast Sikos [2021] and Grojek and Sikos [2022], develop ontology-based frameworks for cybercrime and IoT forensics, allowing semantic enrichment of evidence and more coherent case reconstruction. Although symbolic systems offer high interpretability, they often struggle with scalability as the system complexity grows Le-Phuoc *et al.* [2021]. These limitations highlight the need to combine symbolic strengths with the learning capabilities of ML models.

Researchers have proposed hybrid architectures that combine logic tensor networks and symbolic rules to detect intrusions and respond to cyberattacks. Grov *et al.* [2024]; Jalaian and Bastian [2023]. Their models show improved robustness and transparency compared to conventional black-box detectors. A notable example in the pattern recognition domain is the RuleBoost framework Raza *et al.* [2024], which integrates symbolic constraints into neural models for deepfake detection. These systems illustrate the growing interest in neuro-symbolic AI for applications that demand both high accuracy, auditability, scalability, generalisability or traceability. Nevertheless, to the best of our knowledge, no hybrid AI approaches for digital forensics have explored the use of BDI reasoning to address sensor degradation in IoT environments.

4 Hybrid AI Reasoner

The methodology adopted in this work combines supervised ML with symbolic reasoning to support forensic decision-making in the presence of sensor unreliability. We first describe the digital forensics environment and its characteristics. We then outline the ML workflow, including preprocessing, baseline training, and robustness evaluation under progressive degradation. Next, we introduce the BDI agent architecture, detailing how beliefs, desires, and intentions are operationalised to interact with the ML model. Finally, we explain how we integrate both layers into a unified hybrid AI reasoner that can accept, override, or abstain from predictions, providing an auditable pipeline to support crime investigation in forensic analysis.

4.1 Digital Forensics Environment

The primary dataset for this study, which serves as the basis for the analysis, was originally collected and compiled in Singh *et al.* [2018]. This dataset was collected in a controlled laboratory room. The laboratory comprises a space measuring 6 × 4.6 metres, equipped with four office tables, chairs, a window, and a self-closing glass door. It is important to note that their experiments were conducted with-

out HVAC (heating, ventilation, and air conditioning) systems to maintain consistent conditions during data collection, as forced ventilation could interfere with environmental readings—particularly CO₂ concentration, which tends to decrease under active air circulation.

The laboratory’s network infrastructure uses a Zigbee star topology, with seven slave nodes sending data to a central master node. The use of multiple multivariate nodes was driven by the notion that such a system would offer greater reliability in larger areas than a single-node deployment.

Data were collected at 30-second intervals to ensure thorough capture of environmental changes. This sampling frequency was intended to provide a precise and comprehensive description of the connections between IoT devices and their environment. There are five types of non-invasive sensors: temperature, luminosity, CO₂, sound, and passive infrared (PIR). This diversity of data provides a comprehensive, detailed understanding of the environment under analysis, enabling a more accurate and thorough analysis of occupancy patterns. Table 1 presents the representation of the data that can be found in the dataset. Note that “occupant count” is only used as a ground truth in the process of training an ML model; there is no such sensor in the room. Figure 1 illustrates the spatial context of the data collection process, showing the room layout.

Table 1: Representation of the data in the dataset.

Parameters	Unit	Column Name
Temperature	°C	[S1,S2,S3,S4]_Temp
Luminosity	Lux	[S1,S2,S3,S4]_Light
Sound	Volts	[S1,S2,S3,S4]_Sound
CO ₂	PPM	S5_CO2
Infrared	Binary	[S6,S7]_PIR
CO ₂ Slope	Decimal	S5_CO2_Slope
Occupant Count	Integer	Room_Occupancy_Count

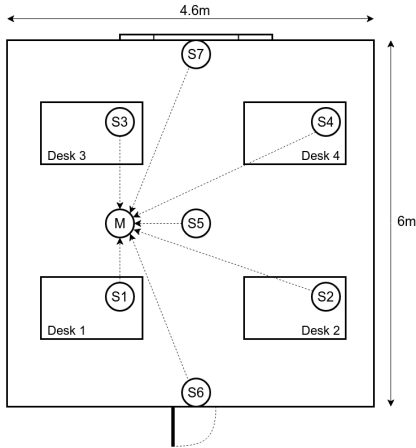


Figure 1: Room layout used to collect data Singh *et al.* [2018].

4.2 Machine Learning Baseline Model

To determine which model would provide the best results for integration with the BDI agent in the digital forensics en-

vironment, we conducted an initial comparison among six standard supervised classifiers applied to the original dataset without any modifications: Decision Tree, Random Forest (RF), K-Nearest Neighbours (KNN), XGBoost, CatBoost, and LightGBM. We trained each model using 80% of the dataset and evaluated on the remaining 20% without any pre-processing or feature scaling, to establish a simple yet consistent benchmark. As shown in Figure 2, all models achieved relatively high accuracy in the clean dataset.

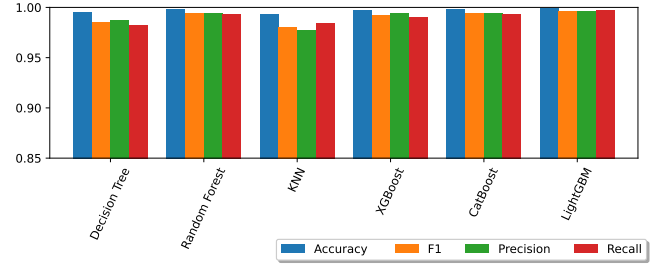


Figure 2: Baseline ML model performances.

Although RF is used as the primary baseline in this work due to its robustness and interpretability, the proposed Hybrid AI reasoner is model-agnostic and can be integrated with any ML classifier that provides confidence estimates. The RF enables the extraction and visualisation of individual decision trees, allowing us to audit and interpret specific model decisions. These characteristics are especially beneficial in forensic analysis, as they provide an intuitive representation of how the model reaches its conclusions and support transparent cross-verification with the symbolic reasoner.

4.3 Data Preprocessing

The dataset contained multiple environmental sensor readings (e.g., CO₂ concentration, light intensity, temperature, and sound levels), together with timestamps and occupancy labels. We encode the target variable as integer classes (0-3) corresponding to occupancy levels.

Preprocessing involved two main steps. First, missing values were imputed using the most frequent value strategy to handle sparse missing data. Second, feature normalisation was applied using Min-Max scaling to rescale all feature values to the [0, 1] interval. This step stabilises model training and prevents attributes with larger numeric ranges (such as CO₂) from dominating those with smaller scales (e.g., sound sensor).

Model Configurations The RF classifier used 400 trees with bootstrap aggregation and Gini impurity. This provided a baseline for occupancy classification and an interpretable ranking of sensor relevance.

Baseline Training and Evaluation The classifier was trained exclusively on clean, non-degraded data to establish baseline references. Performance metrics and feature importance results are reported in Section 5. Unless otherwise noted, all metrics are the result of the $k = 10$ runs with different random seeds. For each seed, we reinitialise the RF

random state, re-sample the stratified train/test split, and reseed the degradation generators, ensuring both the learning and degradation processes vary across runs.

Progressive Sensor Degradation After training, the RF model was evaluated under progressively degraded conditions to assess robustness. Two degradation modes were applied: (i) *Missing Completely at Random* (MCAR), implemented as per-cell random erasure; and (ii) time-windowed sensor outages simulating realistic dropouts. Note that missing data and sensor instability are significant and recurring issues in IoT Forensics Okafor and Delaney [2021]. The MCAR mechanism randomly erases individual sensor readings, simulating transient communication losses, packet corruption, or sporadic sampling failures that occur unpredictably in IoT networks. This random removal of values avoids biasing specific sensors or time intervals, thereby testing the model’s ability to generalise under uncertainty. In forensic scenarios, random missing values represent benign, non-adversarial noise—typical of lossy data acquisition where short communication glitches lead to isolated null Okafor and Delaney [2021]. Time-window outages simulate realistic device-level failures by blanking continuous time segments for particular sensors (e.g., several minutes to hours). These outages replicate scenarios commonly observed in smart environments—such as battery depletion, network disconnections, or deliberate tampering—where sensors become temporarily offline while others continue to operate. This temporal correlation introduces more challenging conditions for the classifier, reflecting the clustered nature of missing evidence often encountered in smart buildings Okafor and Delaney [2021]. Figure 3 shows the temperature *S2_Temp* sensor within a small portion of the testing data, illustrating the combined effect of MCAR degradation and sensor-outage windows applied during the evaluation.

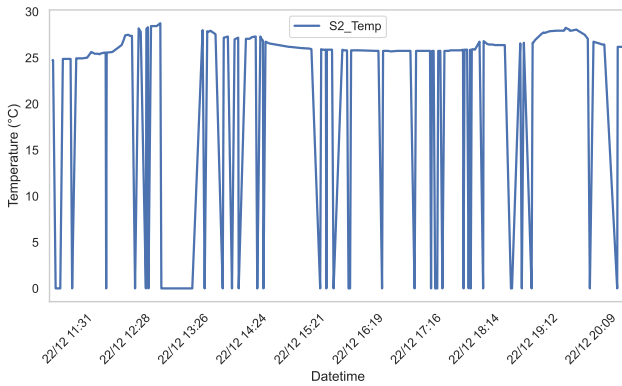


Figure 3: *S2_Temp* sensor with MCAR and time-window outages applied to a small segment of the testing dataset. Zero represents degradation entries (i.e., either through MCAR or time-window outages). While temperature sensors can return zero or negative readings, this was not the case in this dataset.

We chose to represent these degradations using `null` values rather than adding noise to sensor readings. This choice preserves the integrity of the evidence, maintaining the dis-

inction between no observation and faulty observation. Altering sensor values could fabricate evidence, while explicit nulls allow the symbolic BDI agent to reason about the absence of data as a separate source of information. By combining MCAR with structured time-window outages, this approach captures both random missing values and temporally correlated sensor failures, approximating the degradation patterns encountered in real forensic investigations, where incomplete evidence and device unavailability are common due to both operational failures and anti-forensic actions Yaacoub *et al.* [2022]. The resulting dual-degradation mechanism provides a controlled yet realistic basis for evaluating the resilience of our hybrid AI reasoner.

4.4 BDI Agents

The BDI model supports explicit situation-dependent decision policies, allowing the agent to regulate its behaviour based on contextual conditions. For instance, the agent can directly accept high-confidence predictions, fall back to sensor-based reasoning when confidence is low or sensors are unavailable, and apply conservative policies during specific temporal contexts. Another advantage of the BDI model is that it allows decision rules to be modified without retraining the ML model. Each decision emitted by the agent records the triggering plan, the sensor evidence and the ML confidence, producing an auditable reasoning trace aligned with forensic requirements. Furthermore, the BDI agent can be directly integrated into the environment and react to dynamic changes. To implement our symbolic BDI agent, we use the Jason platform for developing autonomous agents Bordini *et al.* [2007]. It builds upon AgentSpeak(L) Rao [1996], a language that formally defines agent programs within the BDI architecture.

At each timestamp, the BDI agent uses the RF prediction and confidence, along with a subset of sensor data (e.g., CO₂, sound, temperature) and contextual factors (e.g., time). Its purpose is to make a single decision on occupancy or abstain from doing so if the information is insufficient.

Beliefs consist of the latest ML output, current sensor readings (which may occasionally be incomplete due to outages and MCAR), and context. The agent’s goal is to deliver an accurate and prompt occupancy estimate. The addition of a timestamp belief triggers the plans that reason about room occupancy. Intentions manifest as exclusive plans; only one plan can activate per timestamp (i.e., plans for reasoning about room occupancy are atomic). Only one plan can succeed per timestamp, and after a decision, the agent prevents additional plans from activating for that timestamp. Each output includes the timestamp, selected class, RF confidence, and plan identifier, facilitating downstream auditing and evaluation for each plan.

To design the BDI agent’s plans, we developed a semi-automated approach for generating sensor heuristics. First, we enumerated all possible combinations of sensor types (CO₂, slope, sound, light, temperature, PIR), yielding 2,500 distinct sensor combinations. Each combination was paired with three simple occupancy heuristics (less-than-or-equal, interval-based, and greater-than-or-equal), with threshold values automatically instantiated using the mean sensor values

observed in the degraded test split.

This process was implemented using Python and SQL scripts, which produced a total of 7,500 candidate plans. From these, we applied an automated filtering step and kept 25 representative plans based on two objective criteria: (i) temporal coverage, measured as the number of distinct timestamps in which a plan could be applied, and (ii) accuracy when correcting low-confidence RF predictions. This selection favours plans that capture recurrent and representative patterns in the data, rather than rules tailored to isolated cases.

Although the plan selection was calibrated using a single reference seed (seed 42), the resulting set of 25 plans was kept fixed and applied unchanged across all experimental runs with different random seeds. This procedure ensures that the symbolic plans are data-driven and reproducible, rather than manually engineered rules.

After designing the BDI plans, we conducted a systematic search to determine the most appropriate confidence threshold for the RF predictions. This threshold dictates when the BDI agent should accept the RF output as valid and when it should intervene with symbolic reasoning. The purpose of the confidence threshold is to balance two objectives: (i) maximising the number of correct predictions accepted directly from the RF, and (ii) ensuring that low-confidence cases trigger symbolic reasoning rather than propagating errors.

4.5 Hybrid Reasoning and Interaction

Our reasoner synthesises ML predictions with symbolic reasoning via a BDI agent system. The ML component is an RF model that infers occupancy from IoT sensor streams. The symbolic layer, implemented as a Jason BDI agent, evaluates the RF output and environmental context to decide when to accept, refine, or abstain from a prediction. This hybrid design aims to preserve accuracy in the presence of missing and unreliable sensors while producing auditable decisions.

At runtime, a Python client streams preprocessed and progressively degraded sensor data to a Java-based BDI server via TCP sockets. Each record contains: a timestamp, degraded sensor values (with `null` indicating missing values), the RF prediction, the RF confidence (defined as the maximum class probability), and optional metadata (e.g., ground truth for evaluation). On the symbolic side, the Jason agent receives these perceptions and reasons over them using explicit plans to (i) accept high-confidence RF predictions; (ii) fall back to sensor-only heuristics when RF evidence is weak or absent; or (iii) abstain when information is insufficient. All decisions are streamed as a single JSON line per timestamp, with the executed plan recorded for auditability.

The symbolic BDI agent continuously monitors the RF model's predictions and the availability of key sensors, serving as an oversight mechanism that applies reasoning to determine room occupancy. We quantify confidence as the average class probability across trees assigned to each instance by the RF model. Based on this confidence and the sensors' contextual reliability, the agent applies a set of plans that determine when to trust the RF model, when to defer, and when to fall back on symbolic reasoning. The sources of decisions are as follows: (i) *High-Confidence Acceptance*: if RF evidence $\geq threshold$, the RF prediction is

directly adopted, with the decision recorded as `source = 'rf'`. This is the first plan in the BDI agent plan library; (ii) *Sensor-Only Heuristics*: if RF evidence is $< threshold$ or unavailable, but informative sensors remain operational, the agent applies domain-specific plans to infer occupancy (`source = 'sensors'`); (iii) *Night Mode*: during 22:00–06:00, conservative plans favour lower occupancy estimates when moderate sound levels are detected, thereby reducing false positives during quiet periods (`source = 'night_mode'`); (iv) *Abstain*: if both RF and critical sensors fail to provide reliable evidence, the agent abstains from making a prediction (`source = 'abstain'`).

The plan `occupancy3p0` is shown in Listing 1 as an example. It encodes a sensor-only heuristic to infer the presence of three occupants when the RF is absent or below the confidence threshold. To ensure deterministic behaviour, the plan is annotated as `[atomic]` and includes the guard `not sent(Dt)`, which ensures that only one plan emits a decision for each timestamp. The plan is triggered by the addition of the belief `datetime(Dt)` and is guarded by three predicates that retrieve the last readings for the sensors: `last_s2_light(Dt, S2_light)`, `last_s3_light(Dt, S3_light)`, and `last_s2_sound(Dt, S2_sound)`. The predicates in the `last_*` format fill in missing values with the latest known measurement when the current timestamp lacks data, thereby improving robustness against short sensor outages.

Listing 1: Jason plan for reasoning about three occupants.

```
@occupancy3p0[atomic]
+datetime(Dt): not sent(Dt) &
  ( last_s2_light(Dt, S2_light) &
    last_s3_light(Dt, S3_light) &
    last_s2_sound(Dt, S2_sound) )
  & ( S2_light >= 98.69 & S3_light >= 133.56 &
      S2_sound >= 0.56 )
<- +sent(Dt);
.concat("\datetime\":", Dt, D1);
.concat(D1, "\",\"bdi_decision\":3, Msg);
java_files.bdi.SendCommand(Msg).
```

Listing 2 shows an example of belief base rules for the S2 light sensor. If a reading is available for that timestamp (Dt), then the value is returned. Otherwise, it retrieves the last available value for S2 light sensor observed before (Dt).

Listing 2: Belief base rules for getting the last s2 light sensor reading available.

```
last_s2_light(Dt, Light) :- s2_Light(Dt, Light).
last_s2_light(Dt, Light) :- not s2_Light(Dt, _) &
  datetime2(Dt2) & s2_Light(Dt2, Light).
```

The decision condition is a conjunction of calibrated thresholds:

$$S2_light \geq 98.69, \quad S3_light \geq 133.56, \quad S2_sound \geq 0.56$$

which were obtained during the systematic exploration of sensor combinations described in Section 4.4. These values represent a state in which illumination and sound indicate the likely presence of three occupants.

Once activated and selected for execution, the body of the plan adds the belief `sent(Dt)` and constructs a structured JSON message with a series of calls to the `.concat` internal action containing the timestamp, the inferred occupancy

Table 2: Mean outcomes across all runs in 10 seeds.

Type	Value
RF Misclassifications	100
Cases Corrected by BDI	69
Cases Actioned by BDI	90
Abstentions (no decision)	0.61%
Overall BDI Accuracy	75.83%
Accuracy of RF Errors Corrected by BDI	68.43%
RF High-Confidence Predictions	89.36%
RF Low-Confidence Predictions	10.64%

level `bdi_decision:3`, and other relevant information (removed from the listing due to space constraints) to facilitate subsequent traceability and cross-verification of decisions.

Operationally, `occupancy3p0` is part of the sensor-only reasoning, which complements other occupancy-specific plans. In practice, `occupancy3p0` is triggered when sensor readings indicate regular activity while the RF model exhibits low or undefined confidence, supplying symbolic corrections under sensor degradation conditions.

5 Evaluation

We evaluated the proposed hybrid AI reasoner in two dimensions: (i) the extent to which the BDI agent contributed by reasoning about room occupancy, and (ii) evaluation of the proposed reasoner’s generalisability with other ML models.

As a baseline, we consider the RF model operating in isolation, without symbolic intervention. This RF-only configuration provides a reference point for quantifying the benefits of our hybrid AI reasoner.

To avoid results based on a single random configuration, we repeated all experiments using ten different random seeds. Each seed reinitialised the RF’s internal state, the stratified train–test split, and the sensor degradation process, generating distinct yet comparable runs. This setup introduced controlled variability in the data partitioning and model behaviour, allowing us to test the system’s consistency under slightly different conditions.

To assess the contribution of symbolic reasoning, we measured how frequently the BDI agent overrode or deferred RF predictions. Table 2 summarises the mean results of 10 runs with all 10 seeds. On average: (i) in 89.36% of cases, the RF prediction was accepted directly (high confidence); (ii) in 10.64% of cases, decisions relied solely on sensor heuristics due to low RF confidence or missing features.

When accounting for these corrections, the hybrid AI reasoner achieved an effective accuracy improvement over the RF-only baseline under degraded conditions. BDI intervened selectively in about 10% of the cases, correcting 68.43% of the RF misclassifications. Across all its interventions, the BDI reached 75.83% accuracy, underscoring its complementary role in low-confidence or degraded conditions while allowing the RF to operate autonomously otherwise.

Table 3 presents the summary per-seed of the BDI agent, showing how often it intervened, how many corrections were successful, and how these results compare with RF errors. The number of interactions between seeds remains fairly consistent, and the BDI regularly corrects a large share of RF

misclassifications. Although the exact counts vary slightly between seeds, the overall result is clear: the BDI contributes meaningful corrections precisely when the RF struggles, without over-activating or introducing unnecessary decisions. These results demonstrate that the hybrid reasoner performs consistently across different random configurations, maintaining a stable and balanced contribution to accuracy.

On average, the agent intervened in about 216 cases per run, resolving roughly 75%. In most seeds, the number of BDI corrections even exceeded the number of RF errors (e.g., seeds 42, 101112, and 192021), showing that symbolic reasoning not only compensates for RF misclassifications, but also promotes more consistent decision patterns overall. The small variation observed across seeds further indicates that the hybrid reasoner remains stable under different random initialisations and degradation scenarios, demonstrating both reliability and robustness.

A closer look at the difference between the number of BDI corrections and RF errors (*BDI Correct – RF Error*) shows a consistently positive margin across all runs, ranging from +35 to +101. This pattern confirms that the symbolic layer adds genuine corrective value, rather than simply mirroring the ML model’s predictions. Even in seeds where the RF had already performed well (e.g., 222324), the BDI maintained a measured level of activation, intervening only when necessary. This behaviour demonstrates that the hybrid architecture effectively adapts to varying model conditions, enhancing overall reliability without introducing redundant decisions.

Table 3: Summary per seed of the BDI agent’s performance.

Seed	BDI Interactions	BDI Correct	BDI Error	RF Error	BDI Correct - RF Error
42	227.0	199.0	28.0	133.0	101.0
123	202.0	149.0	53.0	95.0	57.0
456	233.0	159.0	74.0	133.0	89.0
789	240.0	182.0	58.0	96.0	57.0
101112	236.0	187.0	49.0	115.0	90.0
131415	223.0	163.0	60.0	87.0	62.0
161718	206.0	152.0	54.0	76.0	48.0
192021	228.0	171.0	57.0	123.0	89.0
222324	143.0	108.0	35.0	50.0	35.0
252627	218.0	166.0	52.0	97.0	65.0

To assess the generalisability of the proposed Hybrid AI reasoner, we further evaluated it alongside three other ML models: XGBoost, LightGBM, and CatBoost. Although RF is used throughout the paper as the main baseline for detailed analysis and explanation, this experiment evaluates whether the symbolic reasoning layer remains effective across different underlying ML models. Figure 4 compares the seed accuracy gain achieved by the hybrid AI reasoner over each corresponding ML model across the ten seeds. In all cases, the hybrid approach consistently outperformed the standalone models. The per-seed bars illustrate the variability in outcomes and show how symbolic interventions consistently correct ML misclassifications and improve overall accuracy. Across all ten independent runs for each model, the mean improvement was 0.93 percentage points, with gains observed across all seeds.

6 Discussion

The results indicate that our hybrid AI reasoner provides practical robustness in IoT forensics under degraded sensing

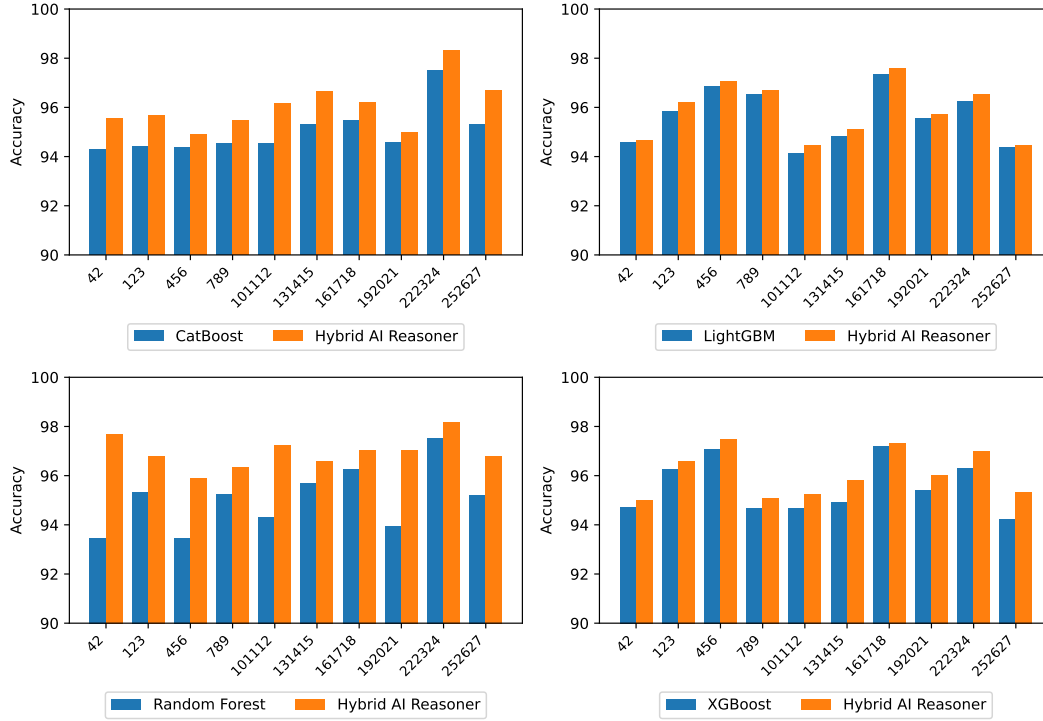


Figure 4: Accuracy (%) for all machine learning models versus hybrid AI reasoner.

conditions. The RF operated autonomously for most cases (89.36% high-confidence accepts), while the BDI agent intervened selectively (10.64%) and abstained rarely (0.61%). Crucially, when intervention was required, the BDI corrected on average 69 of 100 RF misclassifications, achieving 68.43% decision accuracy in the cases it acted upon and 75.83% accuracy across all its decisions. Taken together, these figures demonstrate that the BDI agent makes a significant contribution where the ML model is least reliable.

Although RF is used as the primary baseline, consistent gains with XGBoost, LightGBM, and CatBoost, Figure 4 indicate that the BDI layer functions as a model-agnostic control mechanism. Systematically, the improvement stems from (i) confidence-aware triage (threshold at 0.74) that defers low-certainty RF outputs, (ii) sensor-only plans that remain operative during partial outages, and (iii) conservative, time-aware policies (e.g., night mode) that reduce false positives. Beyond accuracy, the BDI agent adds forensic value. Each decision is auditable (JSON record with plan identifier, timestamp, RF confidence, and sensor state), facilitating the evidential traceability required in investigative settings.

The findings are reported on a single dataset. Generalisability to other buildings, sensor mixes, and temporal regimes remains to be validated. An analysis of threshold choice and plan components, along with calibration of ML confidence, would strengthen causal claims. Finally, sequence-level reasoning (using temporal constraints across windows) was not leveraged and may further reduce residual errors. Our results demonstrate that the BDI agent can act as a low-cost oversight module, enhancing resilience to sensor failures while

preserving an auditable decision trail.

7 Conclusion

This paper introduces a hybrid AI reasoner that integrates ML models with a BDI agent to aid forensic reasoning in smart environments with impaired sensing capabilities. During the evaluation, the ML model functioned independently in most cases, while the BDI selectively intervened when the model's confidence was low, with very few instances of abstention.

These findings illustrate that symbolic reasoning provides substantial improvements where the ML model is weakest. Beyond headline metrics, the contribution is methodological: confidence-informed triage, sensor-exclusive fallbacks, and time-aware policies establish a principled control layer that upholds performance as sensors degrade or shift. Every decision logs its trigger plan, inputs, and model confidence, supporting transparency and post-implementation review, which are essential for digital forensics.

Future work aims to expand the empirical foundation and enhance integration, including: (i) testing on different datasets, particularly considering different buildings, sensor configurations, and degradation scenarios; (ii) exploring learning or adaptive policy techniques (e.g., Bayesian or reinforcement strategies) to decrease manual tuning; (iii) integrating temporal/causal formalisms to reason over sequences and reinforce evidential narratives in comprehensive investigations; and (iv) combining ML explainability techniques (e.g., SHAP Lundberg and Lee [2017]) with explicit symbolic structures of BDI agents, which are inherently interpretable.

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